

1. Problem 1

a.

$$\begin{aligned}\text{function} &= e^{-x/2} \\ \text{operator} &= \frac{d^2}{dx^2} - x^2 \\ \left(\frac{d^2}{dx^2} - x^2 \right) e^{-x/2} &= \frac{e^{-x/2}}{4} - x^2 e^{-x/2}\end{aligned}$$

Not an eigenfunction.

b.

$$\begin{aligned}\text{function} &= \cos(4\phi) \\ \text{operator} &= \frac{d^2}{d\phi^2} \\ \frac{d^2}{d\phi^2} (\cos(4\phi)) &= -16 \cos(4\phi)\end{aligned}$$

This function is an eigenfunction.

eigenvalue = -16

c.

$$\begin{aligned}\text{function} &= \sin \theta \cos \theta \\ \text{operator} &= \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) \\ \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d}{d\theta} \right) (\sin \theta \cos \theta) &= \frac{1}{\sin \theta} \frac{d}{d\theta} (\sin \theta \cos^2 \theta - \sin^2 \theta) \\ &= \frac{1}{\sin \theta} (\cos^3 \theta - 5 \cos \theta \sin^2 \theta)\end{aligned}$$

Not an eigenfunction.

2. Problem 2 (3.1)

$$\begin{aligned}
 \psi &= e^{-bx^2} \\
 \int_{-\infty}^{\infty} \psi^* \psi dx &= \int_{-\infty}^{\infty} e^{-2bx^2} dx \\
 &= 2 \cdot \int_0^{\infty} e^{-2bx^2} dx \\
 \int_0^{\infty} e^{-2bx^2} dx &= \frac{1}{2} \sqrt{\frac{\pi}{2b}} \\
 \int_{-\infty}^{\infty} \psi^* \psi dx &= \sqrt{\frac{\pi}{2b}} \\
 \psi_{\text{norm}} &= ce^{-bx^2} \\
 c^2 \int_{-\infty}^{\infty} e^{-2bx^2} dx &= 1 \\
 c^2 &= \frac{1}{\int_{-\infty}^{\infty} e^{-2bx^2} dx} \\
 c^2 &= \frac{1}{\sqrt{\frac{\pi}{2b}}} \\
 c &= \left(\frac{2b}{\pi} \right)^{1/4} \\
 \boxed{\psi_{\text{norm}} &= \left(\frac{2b}{\pi} \right)^{1/4} e^{-bx^2}}
 \end{aligned}$$

3. Problem 3 (3.5)

$$\begin{aligned}\psi^*\psi &= 4c_1^2 \cos^2(kx) \\ \frac{d(\psi^*\psi)}{dx} &= 4c_1^2 k [-2 \cos(kx) \sin(kx)] \\ &= -8c_2^2 k \cos(kx) \sin(kx)\end{aligned}$$

At maximum

$$\begin{aligned}\frac{d(\psi^*\psi)}{dx} &= 0 \\ -8c_2^2 k \cos(kx) \sin(kx) &= 0 \\ \cos(kx) &= 0 @ kx = \frac{\pi}{2}, \frac{3\pi}{2} \\ \sin(kx) &= 0 @ kx = 0, \pi, 2\pi\end{aligned}$$

Local maxima and minima @:

$$\boxed{kx = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi}$$

4. Problem 4 (3.6)

$$P = \psi^* \psi dV$$

$$P = \psi^* \psi \cdot 4\pi r^2 dr$$

$$\begin{aligned} P &= \frac{1}{\pi a_0^3} \int_0^{a_0} e^{-2r/a_0} \cdot 4\pi r^2 dr \\ &= \frac{4}{a_0^3} \int_0^{a_0} e^{-2r/a_0} r^2 dr \\ &= \frac{4}{a_0^3} \left(\frac{r^2 e^{-2r/a_0}}{-2/a_0} + \frac{2a_0}{2} \int_0^{a_0} r e^{-2r/a_0} dr \right) \\ &= \frac{4}{a_0^3} \left[\frac{r^2 e^{-2r/a_0}}{-2/a_0} + a_0 \left(\frac{a_0 r e^{-2r/a_0}}{-2} + \frac{a_0}{2} \int_0^{a_0} e^{-2r/a_0} dr \right) \right] \\ &= \frac{4}{a_0^3} \left[\frac{r^2 e^{-2r/a_0}}{-2/a_0} + a_0 \left(\frac{a_0 r e^{-2r/a_0}}{-2} + \frac{a_0 - a_0}{2} e^{-2r/a_0} \right) \right]_0^{a_0} \\ &= \frac{-a_0}{2} e^{-2} - \frac{a_0^3}{2} e^{-2} - \frac{a_0^3}{4} e^{-2} + 0 + 0 + \frac{a_0^3}{4} \\ P &= \frac{4}{a_0^3} \left[a_0^3 \left(\frac{-1}{2e^2} - \frac{1}{2e^2} - \frac{1}{4e^2} + \frac{1}{4} \right) \right] \end{aligned}$$

Probability of $r \leq a_0 = 0.323$

5. Problem 5 (3.11)

a.

$$\begin{aligned}
 P_{1/2} &= \int_0^{a/2} \psi^* \psi dx \\
 P_{1/2} &= \frac{2}{a} \int_0^{a/2} \sin^2 \left(\frac{\pi x}{a} \right) dx \\
 P_{1/2} &= \frac{2}{a} \left[\frac{1}{2} x - \frac{1}{4} \cdot \frac{a}{\pi} \sin \left(\frac{2\pi x}{a} \right) \right]_0^{a/2} \\
 P_{1/2} &= \frac{2}{a} \left[\frac{1}{2} \cdot \frac{a}{2} - \frac{1}{4} \cdot \frac{a}{\pi} \sin \left(\frac{2\pi \cdot \frac{a}{2}}{a} \right) \right]_0^{a/2} \\
 \boxed{P_{1/2} = \frac{1}{2}}
 \end{aligned}$$

b.

$$\begin{aligned}
 P_{1/3} &= \int_{a/3}^{2a/3} \psi^* \psi dx \\
 P_{1/3} &= \frac{2}{a} \int_{a/3}^{2a/3} \sin^2 \left(\frac{\pi x}{a} \right) dx \\
 P_{1/3} &= \frac{2}{a} \left[\frac{1}{2} x - \frac{1}{4} \cdot \frac{a}{\pi} \sin \left(\frac{2\pi x}{a} \right) \right]_{a/3}^{2a/3} \\
 P_{1/3} &= \frac{2}{a} \left[\left(\frac{1}{2} \cdot \frac{2a}{3} - \frac{1}{4} \cdot \frac{a}{\pi} \sin \left(\frac{2\pi \cdot \frac{2a}{3}}{a} \right) \right) - \left(\frac{1}{2} \cdot \frac{a}{3} - \frac{1}{4} \cdot \frac{a}{\pi} \sin \left(\frac{2\pi \cdot \frac{a}{3}}{a} \right) \right) \right] \\
 \boxed{P_{1/3} = 0.609}
 \end{aligned}$$

6. Problem 6 (3.12)

$$P(r) = 4\pi r^2 \psi^2$$

$$\psi(r) = \left(\frac{1}{\pi a_0^3} \right)^{1/2} e^{-r/a_0}$$

$$P(r) = 4\pi r^2 \left(\frac{1}{\pi a_0^3} \right) e^{-2r/a_0}$$

$$\frac{dP}{dr} = \frac{4}{a_0^3} \left(2r e^{-2r/a_0} + \frac{-2r^2}{a_0} e^{-2r/a_0} \right)$$

At maximum $P(r)$

$$\frac{dP}{dr} = 0$$

$$0 = \frac{4}{a_0^3} \left(2r e^{-2r/a_0} + \frac{-2r^2}{a_0} e^{-2r/a_0} \right)$$

$$0 = 1 - \frac{r}{a_0}$$

$$\boxed{r = a_0}$$

7. Problem 7 (3.15)

$$\Delta x \Delta p \leq \frac{\hbar}{2}$$

$$v = 5.00 \cdot 10^3 \text{ m/s} \pm 0.15 \text{ m/s}$$

$$\Delta v = 0.3 \text{ m/s}$$

$$\Delta p = m \Delta v$$

$$\Delta p = 9.11 \cdot 10^{-31} \cdot 0.3$$

$$\Delta p = 2.733 \cdot 10^{-31}$$

$$\Delta x = \frac{\frac{\hbar}{2\pi}}{2\Delta p}$$

$$\Delta x = \frac{6.626 \cdot 10^{-34}}{2\pi \cdot 2 \cdot 2.733 \cdot 10^{-31}}$$

$$\boxed{\Delta x = 1.929 \cdot 10^{-4} \text{ m}}$$